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Molecular Insights into the Regulatable Interfacial Property and Flow Behavior of Confined Ionic Liquids in Graphene Nanochannels

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The understanding of confined structure and flow property of ionic liquid (IL) in a nanochannel are essential for the efficient application of ILs in the green chemical processes. In this work, the ionic structure and various flow behaviors of ILs inside graphene nanochannels via molecular dynamics simulations are shown. The effect of the nanochannel structure on confined flow is explored, showing that the width mainly heightens the viscosity while the oxidation degree primarily enhances the interfacial friction coefficient. Tuning the width and oxidation degree of nanochannel, three different flow behaviors including Poiseuille, partial plunger and full plunger flow can be achieved, where the second one does not occur in water or other organic solvents. To describe the special flow behavior, an effective influence extent of the nanochannel (w_{EIE}) is defined, whose value can distinguish the above flows effectively. Based on w_{EIE}, the phase diagrams of flow behavior for the nanochannel structure and pressure gradient are obtained, showing that the critical pressure gradient decreases with width and increases with the oxidation degree. Based on the quantitative relations between confined structures, viscosity, friction coefficient, flow behavior, and nanochannel structure, the intrinsic mechanism of regulating the flow behavior and rational design of nanochannel are finally discussed.

1. Introduction

Ionic liquids (ILs), a new class of solvent that consists of only cations and anions and existing as a liquid near room temperature, have shown many useful properties, such as good thermal stability, nonvolatile, high ionic character, and so on.[1] Due to these unique features, ILs have been sanguinely viewed as new green or environmentally friendly solvents to replace the volatile organic compounds used in the chemical and industrial processes,[2] and have also exhibited many significant applications in the fields of green chemistry, such as high efficiency catalyst,[2b] clean energy storage,[3] treatment of polluted gas,[4] etc.

In the practical application scenario, ILs always combined with other nanoparticles or 2D materials to realize some specific functions such as a high capacity electrode,[5] high selectivity gas separation membrane,[6] safe immobilized catalyst, and so on. The nanoparticles can provide various nanoscale pores and 2D sheets may assemble in paper-like structures with substantial nanochannels. Such confinement can prevent liquids to form bulk structure and has a great influence on the solvation shell of every ion, which plays a key role in many processes, such as ionic transfer and diffusion. Due to the ionic nature and special hydrogen bond (*Z* bond) in ILs,[7] the effect of confinement would be more significant for ILs comparing other conventional molecular liquid, which makes that the confined ILs can have more complex structure and properties.

Inspired by the wide range of ILs in green chemistry, many theoretical simulations[8] and experiments[9] have devoted to understanding the structure and physicochemical properties of ILs under confinement. Feng et al. and Shim et al. found that the distribution of ions and the supercapacitor capacitance of ILs confined in nanopores with varied size showed an oscillatory behavior through the molecular dynamics (MD) simulations.[8c,10] Jiang et al. focused on the electroinduced dewetting and nonequilibrium transport of ILs in bilayer graphene nanochannel and suggested that the ionic current increases sharply as the ion density decrease.[11] Furthermore, the dynamical process of ILs entering or getting out the nanopore and the effect of surface charge and temperature on the ionic diffusion have also been explored via the nuclear magnetic resonance[9e,h] and in situ infrared spectroelectrochemistry.[9c,d] Recently, Futamura et al. found that the Coulombic ordering can be partial breaking and ions of the same charge neighbor each other for ILs confined in a nanopore, namely superionic state, via the X-ray scattering experiment.[8a,9g] Hence, the structure and properties including the interfacial structure,[9g,10] self-diffusion of ions,[12] flow properties under external electric fields[11,13] of confined ILs in nanochannel have been revealed

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systemically. Although the structure and property have been investigated in the prior studies,[14] the quantitative relations between flow properties (especially flow behavior, viscosity, interfacial friction coefficient, etc.) and the nanochannel structure are still unclearly. The limit understanding of the molecular mechanism and quantitative effect of confinement on the flow properties of ILs has restricted the further functional design and application of ILs in the fields of renewable energy and green chemistry.

In this work, large-scale atomistic MD simulations are used to assess the role of confinement on the structure and flow properties of ILs in nanochannel constructed by graphene or graphene oxide. We focus on how the varying width and chemistry of nanochannel affect the interfacial structure of confined ILs and how it further determines nanoflow properties. The atomic structures and self-diffusion property of ILs confined in nanochannel with different width are reported firstly, showing the layered ionic structure near the surface and the restrained diffusive capacity. Then, the intrinsic factors that dominate the flow behavior are identified based on the analysis of interfacial molecular structure and the quantitative flow properties (interfacial friction coefficient, viscosity, slip length, etc.). Moreover, the effective influence extent of the nanochannel is proposed to determine the IL flow behavior inside nanochannel. We found that the criterion derived from the effective influence extent can successfully judge the various flow behaviors including Poiseuille, partial plunger and plunger flows. Finally, the phase diagram of confined IL flow behavior in graphene nanochannel are displayed and the effect of viscosity of different liquid on flow property is also discussed. Our study provides some new insights into the design of nanochannel and control of IL flow behavior, which could benefit the application of IL and graphene in the field of energy storage and conversion or other chemical separator techniques.

2. Results and Discussion

2.1. Atomic Structure and Diffusion of Confined IL in Various Nanochannels

When liquid contacting the substrate surface or confined in the nanochannel, the feature of bulk liquid will be break up and form various nanosized liquid structures. To assess the role of nanoconfinement on the atomic structure of IL, nanochannel was constructed by the 2D graphene sheets as shown in **Figure 1**a. The IL considered in the present work is 1-ethyl-3-methylimidazolium bis[(trifluoromethyl)sulfonyl] imide, namely, $[Emim]^+[TFSI]^-$, as shown in Figure 1b, which has been widely used as the new solvent and electrolyte of

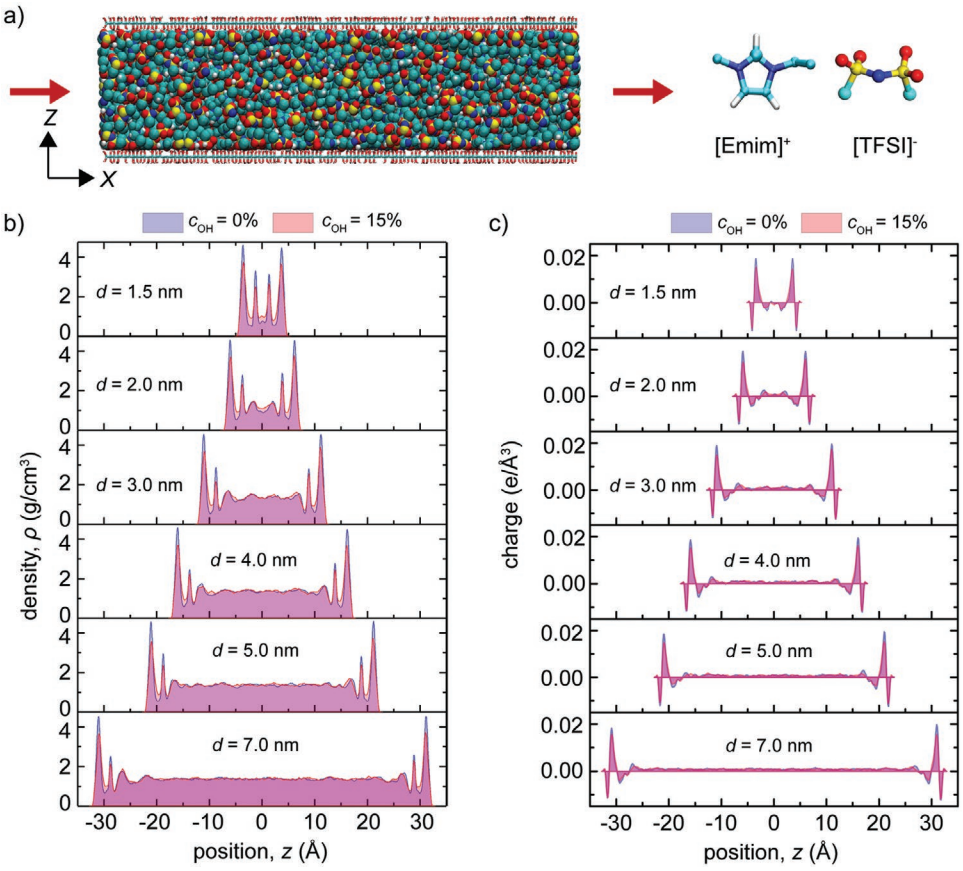


Figure 1. a) The atomic model of IL, $[Emim]^+[TFSI]^-$, and the confined structure for IL inside GO nanochannel, showing an obvious dense layer near the wall surface. The red arrow represents the pressure-driven flow direction. b,c) The distribution of mass density and charge along the nanochannel with width ranging from 1.5 to 7.0 nm, where the blue and red color represents the distribution for graphene and GO with c_{OH} = 15%, respectively.

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supercapacitor and other applications.[9d,e,15] The mass density distribution of confined IL is displayed in Figure 1b, showing that it can form the monolayer, bilayer, and the bulk-like structure as the width of nanochannel (d) increases. For the strong confinement (d < 1.0 nm), the stable monolayer IL structure will form with a relative ordered interaction network while the out-plain interaction is entirely replaced by the surface of the nanochannel. As the intensity of confinement weakening ($d \approx 1.5$ nm), the monolayer IL structure will be destroyed and form a bilayer structure. With increasing the width of nanochannel (d > 2.0 nm), there will emerge the bulk-like IL structure in the region far away from the wall while a dense IL layer remains near the wall. The evolution of confined IL structure with the varied nanochannel is similar to other traditional solvents, such as water, ethanol and so on.[16] The similarity in the layered structure between IL and other organic structure inside nanochannel implies that the unconventional flow feature for traditional solvent may also be applicative for the IL.

The ionic nature of IL makes the charge distribution is also very important for the properties of confined IL. As shown in Figure 1d, we summarized the charge distribution of IL inside nanochannel with d ranging from 1.5 to 7.0 nm. We can see that there will form some charging layer near the wall and the intensity of the charging layer is almost same to different nanochannel, which agrees well with the partial breaking of the Coulombic ordering of $[Emim]^+[TFSI]^-$ via X-ray scattering experiment.[9g] The exist of charging layer or the partial breaking of Coulombic ordering may have a significant effect on the viscosity and flow properties, which will be discussed in the following section. As a comparison, the distribution of density and charge for IL inside graphene nanochannel with c_{OH} = 15% are also displayed in Figure 1b,c. The confined ILs will form hydrogen bond with the oxidative functional groups. Such additive interactions will decrease the intensity of the first peak of density and charge distribution, indicating that some ions ($[TFSI]^-$) will depart from the wall and enter into the adjacent region.

The layered structure of ILs inside nanochannel will restrain the diffusive mobility greatly. As shown in Table S1 of the Supporting Information, the in-plain diffusive coefficients (D_{xy}) for graphene nanochannel with different d are summarized. When the confinement is strong, D_{xy} for $[Emim]^+$ will be constrained to $0.34 \pm 0.001 \times 10^{-11}$ m^2 s^{-1}, only about 5% of the bulk value. As d increases or the confinement weakens, D_{xy} for $[Emim]^+$ will increases gradually and be up to 75% of the bulk value ($5.09 \pm 0.005 \times 10^{-11}$ m^2 s^{-1}). The evolution of D_{xy} for $[TFSI]^-$ have the same trend to that for $[Emim]^+$. The constrained mobility of IL inside nanochannel indicates that the IL flow needs to overcome a larger resistance, especially for the graphene oxide nanochannel. Due to the ionic nature and the complex interactions between IL and graphene oxide (van der Waals interactions, hydrogen bonding, strong electrostatic interactions, etc.), the confined IL flow is expected to have different flow behaviors comparing with the traditional solvent such as water or other organic liquid.

2.2. Different Confined IL Flow Behaviors and Quantitative Interfacial Properties

2.2.1. Three Different Flow Behaviors of Confined IL

To investigate the flow behavior of confined IL, we applied a pressure gradient (P) on the liquid along the nanochannel to realize the pressure-driven flow via nonequilibrium MD (NEMD) simulation as shown in Figure 1a. As shown in **Figure 2**a, the velocity profiles for confined IL in graphene nanochannel

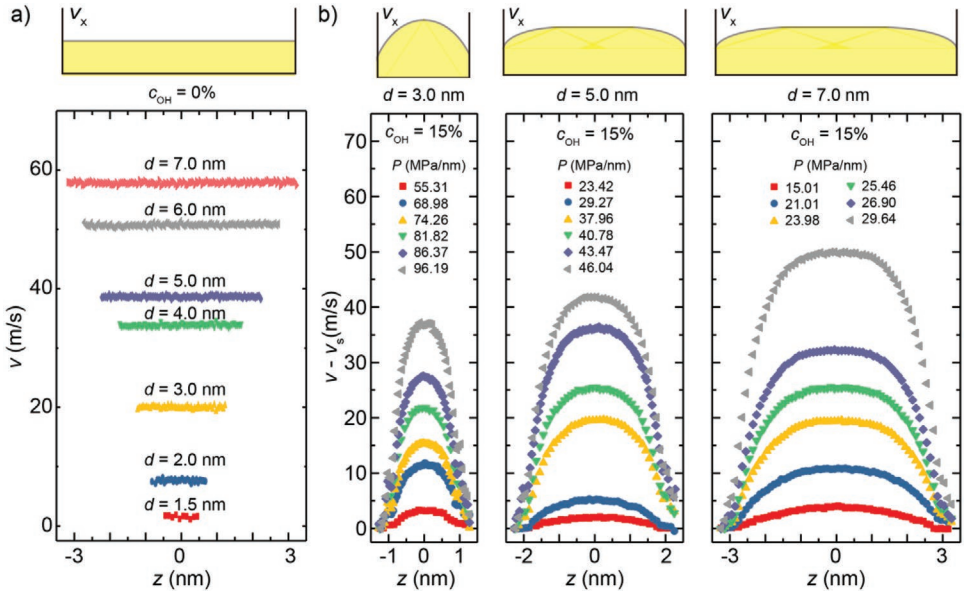


Figure 2. The various nanoflow behavior of confined IL in the nanochannel. a) The velocity profiles of IL confined in graphene nanochannels with width ranging from 1.5 to 7.0 nm, where a pressure gradient of ≈0.21 MPa nm^{-1} is applied. b) The velocity profile of IL inside graphene nanochannel with c_{OH} = 15% for d = 3, 5, and 7 nm, showing the typical Poiseuille flow and partial plunger flow for different d and pressure gradient. The plots in the top panel illustrate the different flow behavior occurred in the confine IL system.

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with d ranging from 1.5 to 7.0 nm were summarized, showing that there exist obviously interfacial slippage for IL inside graphene nanochannel, with notable velocity jump near the surface. For IL inside graphene nanochannel, the full plunger flow can occur due to the atomic smooth wall. Under the same pressure gradient, the average velocity of confined IL will increase with d because of the weaken confinement for the larger channel. When the roughness of the wall is enhanced or oxidative functional groups are added on the wall, the interfacial structure will change significantly as displayed in Figure 1b,c. The velocity profiles for confined IL inside graphene nanochannel with c_{OH} = 15% were showed in Figure 2b and Figure S2 (Supporting Information), indicating that various flow behaviors can occur depending on the width of nanochannel and the external pressure gradient. When the confinement is strong ($d \leq 3.0$ nm), the boundary slippage will be attenuated and the velocity profile meets well with the parabolic curves even if P is very large (≈150 MPa nm^{-1}), showing the typical Poiseuille flow. However, when the size of nanochannel increases ($d \geq 4.0$ nm), the Poiseuille flow will disappear especially for the flow under higher P (Figure S3, Supporting Information). As shown in Figure 2b, the partial plunger flow will occur for d = 5.0 and 7.0 nm, and such flow behavior will be more obvious when d or P increases. As shown in Figure S2 of the Supporting Information, the velocity profile of IL shows a peak velocity near the wall, which arises from the dense layer close to the wall and has been discussed in the recent work.[17] According to the nonlinear number density, we also fitted the velocity profile of ILs in different nanochannels (Figure S4, Supporting Information), indicating that the nonquadratic nature of partial plunger flow is not mainly introduced by the nonlinear density distribution. It should be noted that such partial plunger flow does not apply for other liquid including water or other organic liquid. From the prior work,[16] the fact can be found that water or other organic liquid confined in nanochannel will flow in plunger or Poiseuille form rather than such partial plunger flow. The unique flow behavior should originate the special properties of confined ILs such as high viscosity and complex interaction between IL and graphene oxide.

2.2.2. Interfacial Friction between IL and Wall

To probe the nature of interfacial slip and different flow behaviors, we will first calculate the solid–liquid interfacial friction coefficient λ between IL and the graphene sheet. In the NEMD simulations, we can obtain the interfacial shear stress τ between IL and nanochannel as a function of the slip velocity v_s, which are summarized in **Figure 3**a,b. From the results of τ and v_s for nanochannel with different d and c_{OH}, the fact can be found that τ will increase gradually with v_s in an inverse hyperbolic sine (IHS) relation as $\tau = \tau_0 \operatorname{asinh}(v_s/v_0)$ according to the transistor state theory.[18] Based on the IHS relation, two fitted parameters, τ_0 and v_0, can be obtained, which is related to the friction coefficient via $\lambda = \tau_0/v_0$.[19] The interfacial friction coefficient as a function of the width is displaced

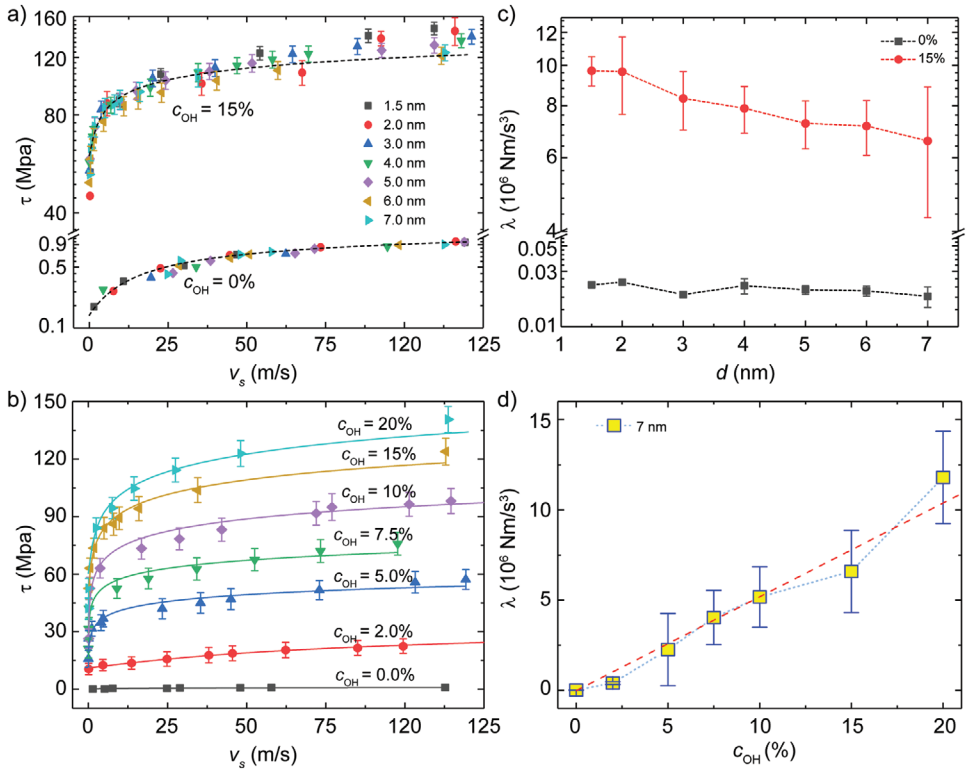


Figure 3. a,b) Friction shear stress τ as a function of the slip velocity at the IL/GO interface for different d and c_{OH}, where dash line represents HIS relation. c) The friction coefficient as a function of nanochannel width for c_{OH} = 0% and 15%. d) The relation between friction coefficient and concentration of hydrogens for d = 7 nm, where the dashed line represents the linear fitting.

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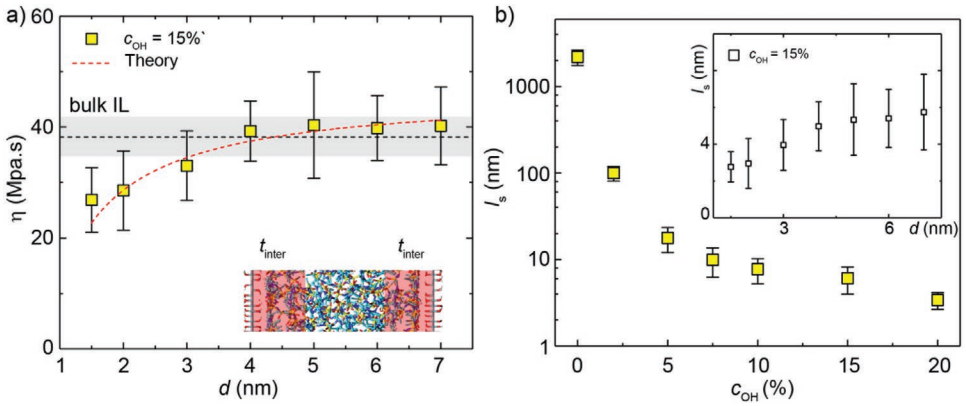


Figure 4. The viscosity and slip length for confined IL. a) The relation between viscosity and width of nanochannel for c_{OH} = 15%, where the gray region represents the viscosity of the bulk liquid. The inset plot shows the interface region of confined IL and the dashed line represents the theoretical prediction for viscosity. b) The slip length for confined IL inside nanochannel with d = 7.0 nm and the inset plot displays the slip length for nanochannel with different size.

in Figure 3c, showing that the confinement has a slight effect on the value of λ. When d increases from 1.5 to 7.0 nm, λ will only decrease about 15.9% and 27.9% for graphene and graphene oxide with 15% hydroxyl, respectively. The slight reduction should originate from the reduction of the confinement on the liquid and agree well with the evolution of confined ILs structure (Figure 1b,c), where the intensity of distribution peak near the wall also changes little. However, the hydroxyls in graphene nanochannel will enhance the friction coefficient greatly due to the introduced massive interactions (Figure 3d). When c_{OH} increases from 0% to 20%, λ will increase from 0.018 to 11.79 × 10^6 Ns m^{-3} linearly (about 650 times enhancement). That's to say, λ is very sensitive to the hydroxyls in the graphene sheet, suggesting that interface engineering can serve as an effective method to tune the confined liquid flow.

2.2.3. Viscous Nature of Confined IL Flow

Besides the interfacial friction coefficient, the confinement can also modify the viscosity of IL, which is the other key factor to describe the flow behavior. The viscosity can be captured using the Green-Kubo equation based on the linear-response theory[19]

$$\eta = \frac{V}{k_B T}\int_0^{\tau_C} < P_{xy}(0)P_{xy}(\tau) > \mathrm{d}\tau \quad (1)$$

where T is the temperature, k_B is the Boltzmann constant, P_{xy} and V are the shear pressure and volume for confined IL at equilibrium, and the angular bracket represents the ensemble average, namely the shear pressure autocorrelation function (ACF).[20] Based on the simulation and ACF of shear pressure shown in Figure S5 of the Supporting Information, the fact can be found that η will increase as τ_C and will be up to convergence at ≈250 ps, so we run at least 10 independent 2 ns long simulations to obtain the average viscosity. It should be noted that the viscosity will also depend on the position of IL in the nanochannel. The IL near the wall may have a larger viscosity due to the dense arrangement of ions while IL in the center region will have a viscosity close to the bulk case. However, the effect of wall on the viscosity of IL at different position is hard to be quantified, hence we will calculate the average viscosity of IL inside the nanochannel via the Green–Kubo method in this work.

The viscosity of bulk $[Emim]^+[TFSI]^-$ from our MD simulation is 38.21 ± 7.06 mPa s, which has a small difference to the experimental value of 32.6 mPa s.[21] However, when IL is placed in the nanochannel, the viscosity will decrease significantly (**Figure 4**a). When d = 1.5 nm, the viscosity will reduce about 30.0% and is lower to 26.86 ± 5.81 mPa s. The reduction should be attributed to the partial breaking of Coulombic ordering of confined IL, especially in the interface region shown in Figure 4a. As shown in Figure 1c, the IL will form several charge layers near the wall and then reduce the Coulombic ordering, which has been proved by Futamura et al. via the X-ray scattering experiments.[9g] Consequently, the electronic interaction within IL will be attenuated effectively and leading to a lower viscosity. When d is up to 5 nm, the viscosity will be close to the bulk value, indicating the reduction depends greatly on the fraction of the interface region. To quantitatively capture the change of viscosity, we can recognize our simulated viscosity via a cross-sectional averaged effective model

$$\eta(d) = (2t_{inter}/d)\eta_0 + (1 - 2t_{inter}/d)\eta_B \quad (2)$$

where t_{inter} is the extent of the interface region, η_0 and η_B are the viscosity for the interface and bulk phase. The effective model can describe the viscosity well as shown in Figure 4a, indicating the rationality of our MD simulations. When the degree of surface oxidation changes, the interfacial structure will have a little change as shown in Figure 1c. Such little change will enhance the viscosity slightly as shown in Figure S6 of the Supporting Information, indicating that viscosity only increases ≈10% when c_{OH} increase from 5% to 20%. It should be noted that the change of viscosity will be more prominent for IL inside small nanochannel, which is because of the higher fraction of interface region for nanochannel with smaller d as shown in Figure 1.

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2.2.4. Slip Length for Confined ILs

As shown in Figure 2, the flow slip is very significant for confined IL, which can be captured by the slip length. The slip length is calculated via $l_s = \eta/\lambda$, here η and λ are the viscosity and friction coefficient from our MD simulations. Figure 4b shows the slip length for IL inside nanochannel $d = 7$ nm with different c_{OH}. For graphene sheet (c_{OH} = 0%), the slip length is about ≈2 µm, which is far large than d of 7.0 nm and lead into a pluglike velocity profile displayed in Figure 2a. For c_{OH} = 5%, the slip length will decrease to ≈17 nm, which is close to the width of the nanochannel. The fact can be found that the interfacial slip will be restrained greatly with the addition of hydroxyls due to the strong interaction between IL and GO sheets. Comparing the sensitive response to hydroxyls on the sheet, slip length increases slightly with the width of nanochannel (the inset plot of Figure 4b). This is because that the IL will form several layered structures and possess a lower viscosity and larger interfacial friction when the confinement is strong and finally leads to a smaller slip length.

The evolution of the viscosity, interfacial friction coefficient, and slip length with size and surface structure of nanochannel indicates that the confined IL flow can be regulated effectively by the rational design of the nanochannel. However, the relation between the interfacial properties and different flow behaviors (Figure 2) is still unclear yet.

2.3. The Intrinsic Mechanism of Various Flow Behaviors

2.3.1. The Nonlocal Model for Partial Plunger Flow and Effective Influence Extent of Channel

As discussed above, the emerge of the partial plunger flow should originate from the balance between interfacial friction, confinement, and viscosity of confined ILs. On the one hand, the interfacial force between nanochannel and ILs will shear the liquid and generate the inner friction within IL, which is the character of the typical Poiseuille flow. On the other hand, the high viscosity for IL will dissipate the shear force quickly and make the IL move as an indivisible whole, which is the feature of the typical plunger flow. When the confinement is strong, the flow is dominated by the interfacial friction and Poiseuille flow occurs; and to the contrary, the flow will be dominated by the intrinsic higher viscosity and partial even full plunger flow realizes. For confined IL inside larger nanochannel, the partial plunger flow occurs through a succession of elastic deformations in the center region and localized plastic rearrangements near the wall.[22] Such flow behavior can be described via the nonlocal model, which introduces a fluidity, $f = (\mathrm{d}\gamma/\mathrm{d}t)/\sigma$, to describe the local flow state. $\mathrm{d}\gamma/\mathrm{d}t$ is the local strain rate and σ is the local stress. The fluidity can be predicted to obey a nonlocal model in the form[22]

$$\sigma = \frac{\gamma}{f} \text{ and } \xi^2 \frac{\partial^2 f}{\partial^2 z} + (f_{\text{bulk}} - f) = 0 \tag{3}$$

where f_{bulk} is the bulk fluidity (without the confinement) and ξ is a variable of the length dimension. Such length ξ describes the influence thickness of the nanochannel on the confined liquid due to the nonlocal elastic relaxation over the entire system.[22] Considering the boundary condition and flow property of bulk IL, we can obtain a solution of Equation (3) and the velocity profile for partial plunger flow of confined IL can be described via the following equation

$$v = \frac{A\xi}{\cosh(d/2\xi)}\left[\begin{array}{c}((z-d/2)\sinh(z/\xi - d/2\xi) - \xi\cosh(z/\xi - d/2\xi)) \\ -((d/2)\sinh(d/2\xi) - \xi\cosh(d/2\xi))\end{array}\right] \tag{4}$$

where A and ξ are the parameters to be fitted, z is the position across the nanochannel and v is the local velocity. As shown in **Figure 5**a, the velocity profile for IL flow inside graphene nanochannel with c_{OH} = 15% can be fitted successfully via Equation (4), where the coefficient of determination (R^2) can be up to 0.99. Then the effective length ξ can be derived from the fitted parameters.

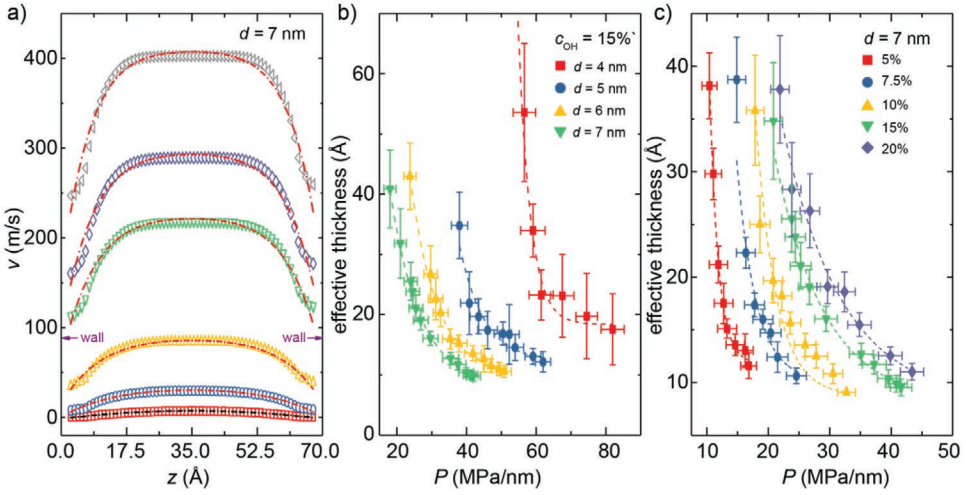


Figure 5. a) The velocity profile for IL inside GO nanochannel with a width of d = 7 nm under different pressure gradient, where the dashed line represents the fitting curve based on the nonlocal model. b,c) The effective thickness derived from the fitting curve of different nanochannel as a function of the pressure gradient for different width and concentration of hydroxyls.

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The effective length for IL flow inside nanochannel with $d = 4.0 - 7.0$ nm is displayed in Figure 5b as a function of the external pressure gradient, showing that ξ will attenuate exponentially to a constant value, which should be attributed to the higher velocity gradient introduced by the higher P. The high velocity gradient will generate a large inner friction within ILs and counteract the increasing shear stress, accelerating the dissipation of friction within ILs and leading to a smaller ξ. The ξ–P relation is also suitable for IL inside nanochannel of 7 nm with $c_{OH} = 5.0 - 20.0\%$ (Figure 5c). The effective length ξ is on the scale of ≈2 nm for ILs in different nanochannels. When ξ is beyond the width of nanochannel, the inner friction will distribute across the entirely confined IL; to the contrary, the region that is far away to the wall will bear almost none inner friction. These two parameters, ξ and d, can both affect the flow behavior, hence, we propose a new parameter, effective influence extent of nanochannel (w_{EIE}), to describe the nanoflow

$$w_{EIE} = 2\xi/d \tag{5}$$

Physically, w_{EIE} describes the spatial spreading of the influence of nanochannel on the confined liquids. The smaller the value of w_{EIE}, the weaker effect on confined liquids can be imposed by the nanochannel. w_{EIE} can be obtained via fitting the velocity profile using Equation (4), hence we transform the effective length to w_{EIE} as shown in Figure S7 of the Supporting Information. From the results, the fact can be found that when P is small or the average velocity is low, w_{EIE} will be larger than 1, showing the Poiseuille flow occurs. However, when P is high, w_{EIE} will be less than 1, indicating the partial plunger flow appears. When the solid–liquid friction is very small (IL in graphene nanochannel), the effective length and w_{EIE} will be close to zero, which means that the emerge of full plunger flow (Figure 2a). That is to say, w_{EIE} can serve as an effective parameter to describe the flow behavior.

2.3.2. The Phase Diagram and Critical Pressure Gradient

To clearly show the phase diagram of flow behavior under different nanochannel and external pressure gradient, the structure-pressure gradient phase diagrams for two conditions are generated. The first case is focusing on the property of the surface while the second one demonstrates the role of nanochannel width in the confined flow. **Figure 6** shows the structure–pressure gradient phase diagrams for both combinations of P versus c_{OH} and P versus d. The results indicate that there exists a critical pressure gradient (P_{cr}) for ILs inside graphene nanochannel: Poiseuille flow will occur when P is less than P_{cr} while partial plunger flow emerges when P is beyond P_{cr}. For the fixed size of nanochannel ($d = 7$ nm for Figure 6a), P_{cr} will increase with the concentration of hydroxyls, where P_{cr} will be close to zero for confined IL inside graphene nanochannel. On the other hand, P_{cr} will decrease as d increases from 4.0 to 7.0 nm (Figure 6b), where partial plunger flow will never occur for $d = 1.5$ to 3 nm even P is up to 150 MPa nm^{-1}. We also analyze the value of w_{EIE} for all the combination of structure and pressure gradient in Figure 5, showing that $w_{EIE} > 1.0$ and $w_{EIE} < 1.0$ are the common feature of the partial plunger flow and Poiseuille flow, respectively.

The flow behavior depends greatly on the interfacial properties of confined IL inside nanochannel. As displayed in Figure 3, the interfacial friction varies obviously with d and c_{OH}, while the viscosity changes little. That is to say, the interfacial friction dominates the flow behavior for IL inside nanochannel. Figure 6c shows clearly the relation between P_{cr} and λ for nanochannel with varied width and concentration of hydroxyls. P_{cr} will increase monotonously with the interfacial friction. From the results, the fact can also be found that the increasing rate of P_{cr} with friction coefficient for IL inside nanochannel of $c_{OH} = 15\%$ is far larger than that for nanochannel of $d = 7$ nm. This is because that IL need to overcome a larger flow resistance in the nanochannel with a smaller width to reach the same velocity comparing with the larger nanochannel. The feature of inner friction (f_{inner}) within IL is also important for the flow process, which is illustrated as a function of width in figure 6d. f_{inner} can be calculated through: $f_{inner} = \eta S(dv/dz)$, where S is the area of IL-nanochannel interface. For the plunger flow ($d = 7$ nm and $P = 0.21$ MPa nm^{-1}), confined IL will move as a whole and f_{inner} will be a constant closing to zero. When Poiseuille flow occurs for IL inside nanochannel with $d = 7$ nm ($c_{OH} = 15\%$, $P < P_{cr}$), the IL will bear a linear distribution of f_{inner} that covers the entire nanochannel, and the velocity profile will show a typical parabolic form. For the partial plunger flow ($c_{OH} = 15\%$, $P > P_{cr}$), f_{inner} will be a constant at the center region that far away from the wall, and the velocity profile will show a nonquadratic nature, where the velocity of ILs in the center region (far away from the wall) will change slightly. From the distribution of f_{inner} of different flow behaviors, the partial plunger flow can be viewed as the combination of Poiseuille flow (interface region) and plunger flow (center region). Hence, we can tune the interfacial friction coefficient to reach these three different flow behaviors, which concurs with the P_{cr}–λ relation shown in Figure 6c. Based on the above quantitative relations between flow behaviors, properties, and the nanochannel structure, the rational design of the nanochannel can be done to control the flow behaviors of IL according to the actual demand from the IL-based applications.

2.4. Additional Remarks

As discussed above, we have shown the quantitative relation between flow behaviors and width, surface defect or interfacial friction coefficient. However, the relation between flow behavior and viscosity is absent now because the viscosity is hard to control through width and surface defects (Figure 4). In the recent work, Jiao et al.[16b] studied the confined structures and nanoflow of seven solvents (acetone, hexane, toluene, water, ethanol, butanol, and ethylene glycol) and found that there existed only two flow behaviors: plunger flow for graphene channel and Poiseuille flow for graphene oxide channel. The viscosities for the above solvents are summarized in Table S2 of the Supporting Information, which range from 0.254 to 10.666 mPa s and are far less than that for the IL considered in the present work. Hence, the viscosity should be a necessary condition for the emerge of partial

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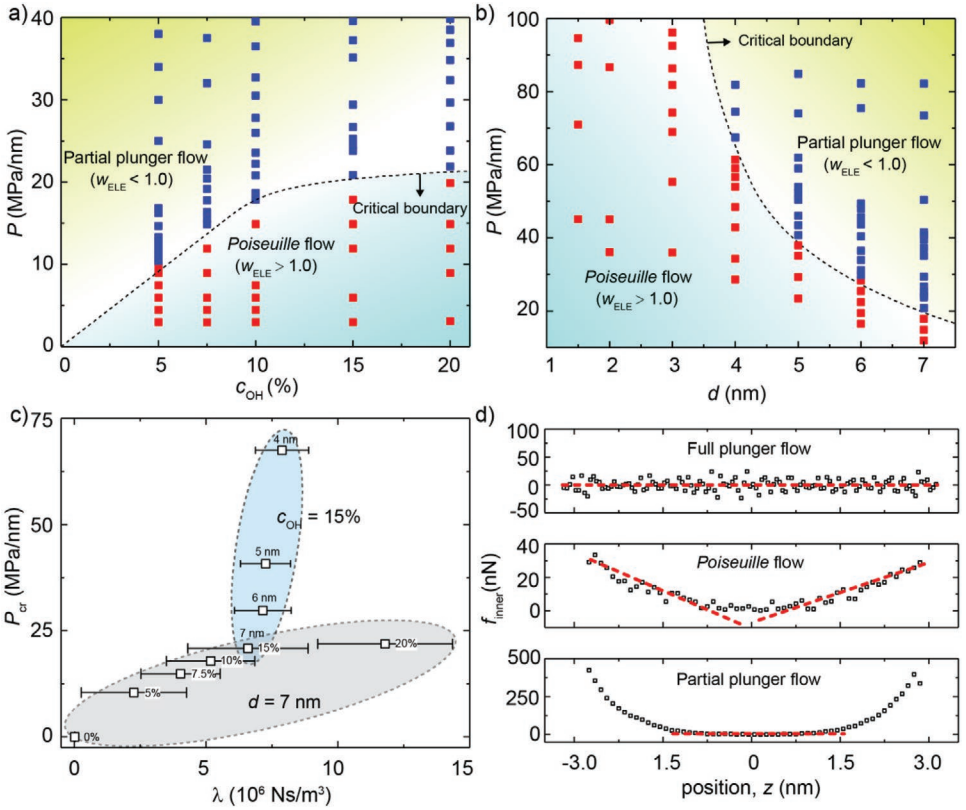


Figure 6. The simulated results of Poiseuille flow (red square) and partial plunger flow (blue square). a) The surface hydroxyl concentration-pressure gradient phase diagram for d = 7 nm. b) The nanochannel width-pressure gradient phase diagram for c_{OH} = 15%. The flow behavior agrees well with the value of w_{EIE}: w_{EIE} > 1.0 and w_{EIE} < 1.0 representing Poiseuille and partial plunger flow, respectively. c) The relation between the critical pressure gradient and the interfacial friction coefficient for IL inside different nanochannel. d) The different distribution of the inner friction within IL for three different flow behaviors. The data for full plunger flow, partial plunger flow, and Poiseuille flow is derived based on the velocity profile of IL inside nanochannel of d = 7 nm with c_{OH} = 0% (P = 0.21 MPa nm^{-1}), c_{OH} = 15% ($P < P_{cr}$), and c_{OH} = 15% ($P > P_{cr}$), respectively.

plunger flow. When the viscosity is low, the generated inner friction is small and the liquid can be shear easily, which will make liquid flow in the Poiseuille form. To the contrary, the higher viscosity will make the liquid move as a whole and only the interface will be affected, where liquid will prefer to flow in the partial plunger form. For the graphene channel, all liquid with different viscosities ($0.25 < \eta < 108.94$ mPa s) will flow in a plunger form due to the little friction between liquid and atomically smooth graphene.[16b,19] However, the quantitative relation between viscosity and flow behaviors, especially the critical viscosity to reach different flow behaviors for liquid in a specific nanochannel, requires further studies.

3. Conclusion

In brief, we explore the ionic structures, flow properties and behaviors of $[Emim]^+[TFSI]^-$ IL via massive MD simulations, which is confined within 2D nanochannel constructed by graphene and graphene oxide sheets. It is found that IL will form a dense layer near the surface, and the confined structure and properties can be tuned effectively by two different methods: the width and chemistry of the nanochannel. The width mainly heightens the viscosity of confined IL while the effect on the interfacial friction coefficient is weak. Contrarily, the chemistry will enhance the interfacial friction while affecting the viscosity slightly. Inside the nanochannel with different width and chemistry, IL can reach three different flow behaviors including Poiseuille flow, partial plunger flow and fully plunger flow. To describe the different IL flow behaviors, an effective influence extent of the wall is derived from the nonlocal flow model, which also depends greatly on the structure of nanochannel. Based on the effective influence extent, a critical pressure gradient, P_{cr}, can be determined, where the Poiseuille and partial flow will be reached for pressure gradient being less or beyond P_{cr}, respectively. Furthermore, the phase diagrams for IL flow behavior are obtained, showing that the structure of nanochannel mainly affects the IL flow behavior and P_{cr} through the interfacial friction coefficient. Finally, the effect of viscosity of different liquid on the flow behaviors and potential design of nanochannel for the practical applications are discussed. These results and understandings on the confined IL and flow behaviors can be extended to the nanopore or nanochannel constructed by other materials and could help the rational design and management of IL-based applications in the field of gas separation or other key green chemical processes.

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4. Simulation Section

4.1. Atomic Structure

The surface oxidation of graphene sheet can be performed via adding different functional groups on the graphene layer such as hydroxyl, epoxy, and carbonyl, and so on,[23] where hydroxyl groups remain rich and toward a long-living quasi-equilibrium state. Hence, we only consider the hydroxyls groups in the structure of graphene oxide. The concentration of these defects is defined as $c_{OH} = n_{OH}/n_C$, where n_{OH} and n_C are the number of the oxidative atom site and the total number of carbon atom in the pristine graphene sheet. The distribution of hydroxyls in graphene is sampled randomly as shown in Figure S1a of the Supporting Information. The graphene sheet prefers to self-assemble in paper-like structures with interlayer distance on a nanometer scale and construct many nanochannels. In this work, we will place two graphene sheets parallel and construct a graphene nanochannel with d = 1.5 to 7.0 nm. It should be noted that the actual nanochannel constructed by 2D materials will be very complex (such as wrinkle surface, inhomogeneity size distribution, different topology nanochannel, etc.), here we only consider the ideal case (Figure 1) to simplify the simulations. To determine the number of confined ILs in the graphene nanochannel with d, we insert a graphene bilayer with d into the box of pre-equilibrated ILs. Here, the typical IL, $[Emim]^+[TFSI]^-$, namely, 1-ethyl-3-methylimidazolium bis[(trifluoromethyl)sulfonyl]imide, is considered due to its wide utilization in the field of gas separation, supercapacitor, and so on.[9d,e,15] After a long enough MD simulation ≈10 ns), the number of IL confined in graphene bilayer will be convergent. Varying the distance, a series of number N_{IL} can be obtained via the same approach, which is displayed in Table S3 of the Supporting Information. Then the atomic structure shown in Figure 1a and Figure S1b–d (Supporting Information) will be constructed, where the size of the graphene sheet is 5.40×15.00 nm^2. The periodic boundary condition is applied in the x- and y-directions while an open boundary condition is adopted in the z-direction.

4.2. MD Simulations

All the simulations in this work are performed via large-scale atomic/molecular massively parallel simulator.[24] The timestep for integrating Newtonian equations of motion is 1.0 fs, which has been validated to ensure energy conservation. The all-atom optimized potential for liquid simulations (OPLS-AA) was employed for graphene and graphene oxide,[25] which has been used successfully to predict the structural and mechanical properties of graphene oxide.[16a,23b,26] Recent work has shown that the nonpolarizable force field can also predict a reasonable interfacial structure at IL/graphene interface via the ab initio MD simulation,[27,28] although such force field may ignore the effect of molecular polarizability.[29] Such a force field can also be used successfully to calculate the self-diffusive constant, viscosity, nanoflow behavior, and so on.[30] Hence, we adopt the OPLS-AA force field to describe the van der Walls interactions, electrostatic interactions, bond, angle, and dihedrals in ILs. The interactions between graphene and IL include both electrostatic and van der Waals terms. The former one, long-range Coulombic interaction, is computed using the particle–particle–particle–mesh algorithm.[31] The later one is described using the Lennard–Jones potential with a cutoff of 1.2 nm, and the Lorentz–Berthelot mixing rules are used to model the parameters of graphene and IL. The SHAKE algorithm is also used to reduce high-frequency vibrations of hydrogen in graphene oxide.[32] In the simulations, the carbon atoms in the graphene sheet are fixed to maintain the planar structure of the sheet considering 2D sheet staked in a thin film case in the paper-like membrane.[6] The oxygen functional groups and ILs are kept at the temperature of 300 K using the Berendsen thermostat,[33] with the temperature calculated after removing the center-of-mass velocity.

4.3. Simulation of the Pressure-Driven Nanoflow

To produce the pressure-driven flow, the external force is added to every atom in IL as shown in Figure 1b. After adding the external force, the ions will move along the nanochannel and will archieve the equilibrium state after a longer enough MD simulations (balance between external force, friction from nanochannel, and inner friction within ILs). The process always needs to spend about 5–10 ns depending on the width or surface oxidation degree of the nanochannel. After the steady flow is archieved, an additional 2 ns MD simulation will be performed to collect the data for subsequent analysis, which is long enough to obtain the convergent velocity as shown in Figure S3 of the Supporting Information (compared with the 20 ns long simulations). The distribution of temperature of ILs in flow and velocity profile of flow using a different thermostat, showing an unbiased-temperature-profile and indicating that the thermostat should have a litter effect on the dynamics of the pressure-driven flow.[34] Such nonequilibrium MD simulations have been implemented well in the recent studies on liquid flow in the nanochannel.[16a,19,35] Through applying different external forces, the various flow behaviors can be obtained as shown in Figure 2 and Figure S2 (Supporting Information).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

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Keywords

graphene nanochannels, ionic liquids, molecular dynamics simulations, nanofluids, 2D materials

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